Image-based Trajectory Determination of Artificial Satellites

State of the art, critical assessment and conclusion

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1 Introduction

Previously, the preliminary paper established underlying geometric fundamentals and four methods for orbit determination: two based on a single-camera and two on a dual-camera setup, with one method in each setup determining circular orbits and the other full orbital parameters. The research paper continued with experimental setups to evaluate how these methods react to variable parameters, star-based calibration and different satellite altitudes. The aim of this concluding paper is to contextualize results into the state of art and assess accuracies, in case that comparable metrics are provided. Additionally, certain issues of methods need to be addressed. Lastly, we aim to conclude which methods retain their practical use-case in the scope of image-based orbit determination with short observation arcs.

2 State of the art

The following section provides an exemplary overview of two ground-based, non-imaging methods to illustrate how the approaches presented in our previous work [3, 4] relate to the broader field of satellite orbit determination.

Laser-based:
In this approach, a laser beam is directed at the satellite, which reflects light back towards the observer. Distance between observer and satellite is determined by measuring time, which the light takes to reach and return. Bamann et al. present a setup where this method is tested on space debris. Similar to the progression error f_p, as established in our research paper, they compare absolute distance between object on determined orbit and a supposed ground truth, which yields a drift of up to 450m. [1]

Radar-based:
In this approach, as presented by Hough, three ground-based radar units emit non-directed radio pulses synchronously. The satellite reflects them back to their senders, where time between send and return is measured. Through means of trilateration, absolute satellite positions are obtained with a precision of less than 10m. These positions are then used for determining orbital parameters. [2]

3 Critical assessment

3.1 Observation difficulties in real world settings

Previously, the research paper established an experimental setup on how the orbit determination methods can be subjected to variable parameters to evaluate their accuracy under

different conditions. Results indicate, that methods which do not compromise on a circular model, require long observation arcs $\Delta\theta$ or Δt [4] as pictured below in figure 1.

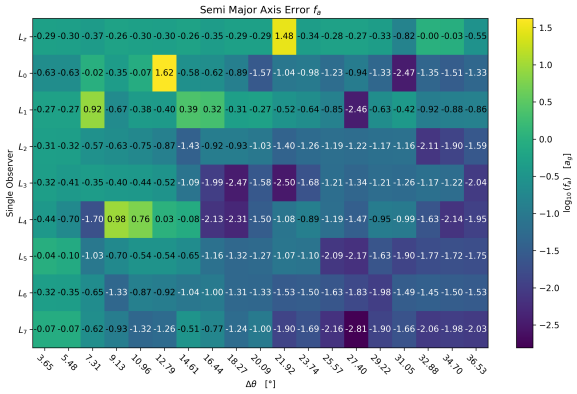


Figure 1: Gaussian method requiring large $\Delta\theta$ for reasonable parameter prediction [4].

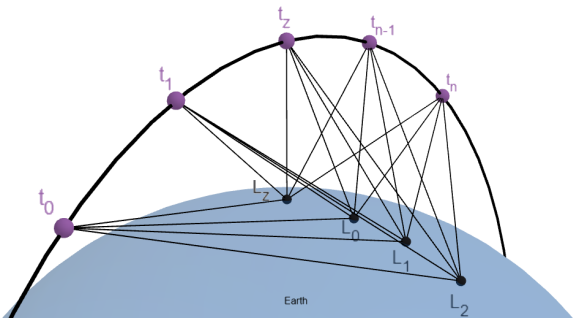


Figure 2: Schematic of strategic camera placement [4].

Figure 2 illustrates the consequences of long observation arcs (e.g. from t_0 until t_n), which poses some difficulties in real world settings. For one, observations close above the observer's horizon might be inaccurate as the lines of sight experience more severe refraction effects of Earth's atmosphere. Secondly, continuous observation requires clear skies and unobstructed lines of sight throughout the entire passage, imposing restrictive requirements on observer locations and prevailing weather conditions. Lastly, time frames of the day when satellite observation becomes feasible is at dusk or dawn, as satellites are illuminated by the sun, not covered by Earth's shadow [5] and scattered sunlight is low enough for them to be visible from the ground.
Overall, several restrictive requirements regarding observation time and location can not be omitted. Observations close to the horizon may be infeasible due to refraction effects.

3.2 Numerical instability of the Gaussian method

Mentioned previously, the research paper suggests that the Gaussian method contains a numerical instability for observations from observer location L_z or close [4] as visible in figure 1, where semi major axes tend to be predicted more reliably, the further the observer is located from the orbital plane. After close inspection, this issue seems to arise from the D_0 scalar, or rather a set of equations as part of the algorithm, where

$$D_0 = v_1 \cdot m_1 \quad \text{with} \quad m_1 = v_2 \times v_3$$

gets inserted into

$$A = \frac{1}{D_0}\left(-D_{12}\frac{\tau_3}{\tau} + D_{22} + D_{32}\frac{\tau_1}{\tau}\right)$$

$$B = \frac{1}{6D_0}\left(D_{12}\left(\tau_3^2 - \tau^2\right)\frac{\tau_3}{\tau} + D_{32}\left(\tau^2 - \tau_1^2\right)\frac{\tau_1}{\tau}\right)$$

[6]
This proximity to the orbital plane causes view vectors v_1, v_2 and v_3 to lie approximately within a plane, resulting in D_0 to approach zero. Thereby, A and B diverge to large, imprecise values due to floating-point division by D_0.
Consequently, to be able to perform orbit determination, observers must not be placed, so

that the satellite passes in their zenith. This aspect is sub optimal for orbit determination of low altitude orbits because placing an observer further from the orbital plane interferes with the ability to track the satellite over longer observation arcs $\Delta\theta$.

3.3 Weakness of the simplified Gaussian method

The simplified Gaussian method assumes a circular orbit by default. Research results have shown, that it is quite accurate in predicting a satellite's trajectory within seconds if the underlying ground truth orbit is also circular [4]. Issues arise if the underlying ground truth orbit is elliptical which is investigated in this section. First, we recapitulate one of the initial ideas it is based upon, namely the relation between recorded time stamps and orbital period:

$$t_2 - t_1 = \Delta t = \frac{\varphi}{2\pi} T \tag{1}$$

Hereby, φ equates the angle of an arc which the satellite traverses between t_1 and t_2. [3]
In other places we use $\Delta\theta$ to describe the passage angle. We insert $\Delta\theta$ for φ and reformulate equation 1 as follows:

$$\Delta t = \frac{\Delta\theta}{2\pi} T \Leftrightarrow \frac{2\pi}{T} = \frac{\Delta\theta}{\Delta t} = \omega$$

We obtain $\frac{\Delta\theta}{\Delta t}$, an angular velocity value ω (not to be confused with argument of periapsis). Furthermore, it is trivially known that circular trajectory velocity v relates to angular velocity in scalar-notation as follows:

$$v = \omega \cdot r_\perp \Leftrightarrow \omega = \frac{v}{r_\perp} \tag{2}$$

This relation holds if the velocity vector is perpendicular to the radius vector, thus at every position for circular trajectories. Regarding elliptical trajectories, this relation without modification only holds at periapsis and apoapsis from a ground-based observer's point of view, which is sufficient for our purposes.
Next, Weber provides the *Vis Viva* equation, which describes current velocity v subject to current radius r of an object moving along an elliptical trajectory with given semi major axis a as follows:

$$v = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)} \tag{3}$$

[7]
This equation can be reformulated as angular velocity

$$\omega = \frac{\sqrt{\mu \left(\frac{2}{r_\perp} - \frac{1}{a} \right)}}{r_\perp} \tag{4}$$

which holds for $r_\perp \in \{a, r_{pe}, r_{ap}\}$. Suppose that the simplified Gaussian method is applied to a satellite on an elliptical trajectory, located at its trajectory's periapsis or apoapsis, we can now determine what semi major axis a_m will be predicted from ground truth semi major axis a_g subject to eccentricity $0 \leq e < 1$.
The approach is to equate expression 4 with itself for $r_\perp = r_{pe/ap}$, $a = a_g$ and $r_\perp = a_m$, $a = a_m$ and rephrase it to obtain a_m. Hence:

$$\frac{\sqrt{\mu\left(\frac{2}{r_{pe/ap}}-\frac{1}{a_g}\right)}}{r_{pe/ap}}=\frac{\sqrt{\mu\left(\frac{2}{a_m}-\frac{1}{a_m}\right)}}{a_m}\Leftrightarrow$$

$$\frac{\sqrt{\left(\frac{2}{r_{pe/ap}}-\frac{1}{a_g}\right)}}{r_{pe/ap}}=(a_m^{-1})^{\frac{1}{2}}\cdot a_m^{-1}=a_m^{-\frac{3}{2}}\Leftrightarrow$$

$$\left(\frac{\sqrt{\left(\frac{2}{r_{pe/ap}}-\frac{1}{a_g}\right)}}{r_{pe/ap}}\right)^{-\frac{2}{3}}=a_m$$

It is trivially known that $r_{pe/ap}=a(1\mp e)$, hence we insert this term to obtain $a_{m(pe/ap)}$ subject to e:

$$a_{m(pe/ap)}=\left(\frac{\sqrt{\left(\frac{2}{a_g(1\mp e)}-\frac{1}{a_g}\right)}}{a_g(1\mp e)}\right)^{-\frac{2}{3}}=\frac{a_g(1\mp e)}{\sqrt[3]{1\pm e}}$$

Thus:

$$\boxed{a_{m(pe/ap)}=\frac{a_g(1\mp e)}{\sqrt[3]{1\pm e}}} \tag{5}$$

This formula yields the values which the simplified Gaussian method would determine for measurements at periapsis and apoapsis. It works under the assumption that Δt is short, meaning no significant change in ω between recordings.

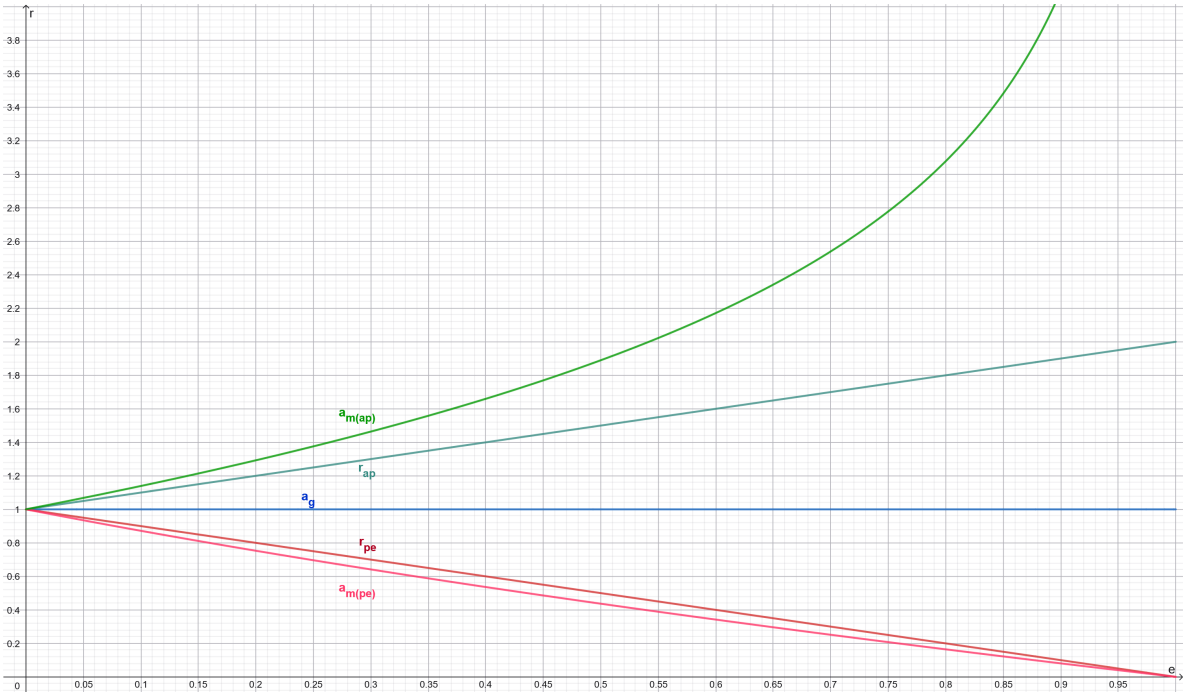


Figure 3: Predicted semi major axis values a_m if measurements are conducted at periapsis or apoapsis subject to eccentricity $0 \leq e < 1$.

Pictured above are curves of periapsis and apoapsis radii r_{pe} and r_{ap} and predicted semi major axes $a_{m(pe)}$ and $a_{m(ap)}$ subject to eccentricity $0 \leq e < 1$ for generic ground truth semi major axis $a_g = 1$ as output of equation 5.
We observe that, when measured and applied at periapsis, predicted semi major axes lie even below r_{pe}, meaning that the predicted circular orbits would lie completely within the elliptical ground truth orbit. As for measurements at apoapsis, we observe that predicted semi major axes lie above r_{ap} with growing tendency as e increases. This means that the ground truth elliptical orbit lies completely within predicted circular orbits.
Predicted semi major axes from other positions than periapsis or apoapsis, which is the average case, would lie in between these values and might be a closer match to a_g. However, this approach shows how far predictions might deviate from reality under poor circumstances, highlighting the potential instability of the simplified Gaussian method, rendering it suitable for near-perfect circular orbits only.

4 Conclusion

On the whole, all four presented image-based methods in the scope of short observation arcs yield less precise results than laser- or radar-based approaches, with propagation and semi major axis errors f_p and f_a being in the range of a few kilometers at best. Hereby, it must be noted, that methods for full parametric orbit determination yield considerably worse results than methods assuming circular orbits. Considering observation issues in real world settings, where the maximum achievable observation span $\Delta\theta$ might actually be shorter than 1/10 of an arc due to refraction effects near the horizon, full parametric orbit determination can only be considered to be a rough initial guess at best. To obtain better results using the Gaussian method or stereo vision for full determination, more precise equipment and/or a different camera setting may be needed, with cameras placed in locations around the world. The claimed 3/4 of an arc to several resolutions [8] are a more reasonable guess.
The great advantage of the simplified Gaussian method and stereo vision for circular determination are truly short $\Delta\theta$ in the range of 1/1000 of an arc required for determining parameters. Hereby, the stereo vision method possesses the advantage that it is not affected by the issue described in section 3.3. The disadvantage compared to simplified Gauss is that it needs an additional camera. If to-be predicted orbits are truly circular, these methods might yield quite accurate parameters, although not suited for high-precision applications.

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